

Application of Neural Network to Brain-Computer Interface

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Abstract

In this study, an neural-network-based system is proposed for the applications of brain-computer interface (BCI). Applying event-related brain potential (ERP) data acquired from the sensorimotor cortices, the system consists of three procedures, including enhanced active segment selection, feature extraction, and classification. Firstly, combined with the use of continuous wavelet transform (CWT) and Student's two-sample t-statistics, the 2D anisotropic Gaussian filter is proposed to further refine the active-segment selection. Multiresolution fractal features are then extracted from wavelet data by means of modified fractal dimension. Finally, support vector machine (SVM) is used for classification. Compared with other approaches on motor imagery data, the results indicate that the proposed method is promising in BCI applications.

Keywords: Brain-computer interface (BCI); Electroencephalogram (EEG); Active segment selection; Wavelet transform; Modified fractal dimension; Support vector machine (SVM).

1. Introduction

The objective of brain-computer interface (BCI) is to provide human beings with an alternative communication channel for directly transmitting messages from the human brain to computers by analyzing the brain's mental activities [1, 2]. The BCI is described that a person has the ability to communicate with others without the prerequisite of brain's normal output pathways of peripheral nerves and muscles by controlling his EEG signals [3]. BCI systems based on the single-trial analysis of electroencephalographic (EEG) signals associated with motor imagery (MI) have grown rapidly in the last decade [1]. The principal aim of this study is to

propose an neural-network-based BCI system for MI EEG data classification.

Continuous wavelet transform (CWT) is applied for the precise localization of ERP components since it gives a highly complete representation of EEG signals in the time-scale domain [4]. It is somewhat time consuming, so the CWT and Student's two-sample t-statistics are used to off-line select active segments in the time-frequency domain. In addition, in order to further refine the location of active segments, the 2D anisotropic Gaussian filter is proposed to make the curve of profile smoother. Feature extraction is also an important issue in BCI work that substantially affects the classification accuracy. An effective feature extraction method can help increase classification performance. A variety of feature extraction methods has been proposed. Among them, the band power and AAR parameter model are popular and commonly used [5–8]. Feature extraction based on band power estimation is usually obtained by computing their powers at several alpha and/or beta bands. The estimated band powers are then computed with their logarithm values as descriptive parameters for every channel [5], or estimated by averaging over them [6]. On the other hand, AAR models are also popular for feature extraction in mental tasks [7, 8]. The EEG time series is fitted with an AAR model.

Moreover, fractal geometry [9] provides a proper mathematical model to describe complex and irregular shapes that exist in nature by using fractal features. We propose modified fractal dimension to extract multiresolution fractal features from wavelet data for subsequent classification. As the DWT is powerful in selecting features in multiresolution, it is an efficient and structured approach in ERP representation [10, 11]. Fractal dimension is a common method for the extraction of fractal features. It has been applied to various biomedical signal analyses. The multiresolution fractal feature vectors (MFFVs) contain not only multiscale characteristics but important fractal

information. Hence, based on multi-resolution concepts, we provide a solution to describe the characteristic of fractal with different wavelet scales by MFFVs [12].

Finally, support vector machine (SVM) is used for classification in this study, since it has a very steady theory in statistical learning and guarantees to obtain the optimal decision function from a set of training data.

2. Materials and methods

A flowchart of EEG analysis system for single-trial classification of MI data is illustrated in Fig. 1. It mainly consists of three procedures: enhanced active segment selection, feature extraction, and classification. Enhanced active segment selection further refines the location of active segments with the 2D anisotropic Gaussian filter. The multiresolution fractal features are then extracted from wavelet data by proposed modified fractal dimensions. Finally, the SVM classifier is used for the discriminant of features.

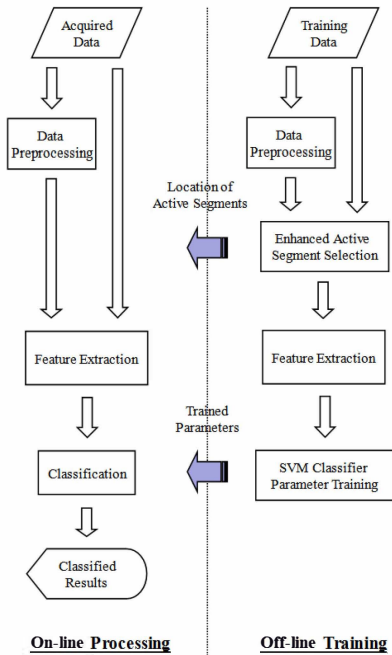


Fig. 1. Flowchart of the proposed BCI system. It contains three procedures: enhanced active segment selection, feature extraction and classification.

2.1. Materials

A dataset, recorded by the Graz BCI group, is used to evaluate the performance of proposed system in the experiments. There were three subjects during a feedback experimental recording procedure. The task

was to control a bar by means of imagery left or right hand movements [8, 13, 14]. The order of left and right cues was random. The data was recorded on three subjects – the first subject, S1, performed 280 trials, while other two, S2 and S3, each performed 320 trials. The length of each trial was 8–9s. The first 2s of the trial were quiet, then an acoustic stimulus indicated the beginning of the experiment at $t = 2s$, and a fixation cross + was displayed for 1s; then, at $t = 3s$, an arrow (left or right) was displayed as a cue. (Only the data recorded between 3 and 8s were therefore considered to be event-related.) At the same time, each subject was asked to move a bar by imagining left or right hand movements according to the direction of the cue. The recordings were made using a g.tec amplifier and Ag/AgCl electrodes; all signals were sampled at 128 Hz and filtered between 0.5 and 30 Hz.

2.2. Enhanced active segment selection

An event-related window of length 4-s or 5-s is too long prior to feature extraction. It is important to reduce the event-related window to 1-s active segment, which is defined as the most discriminative from the event-related window, to achieve efficient computation and high classification accuracy. The CWT [4] is applied for precise localization of the ERP components in the time-frequency domain. More specifically, the CWT and Student's two-sample t-statistics are performed to select active segments from the time-frequency domain. The CWTs of EEG signals for each trial, which is of either left or right data, are analyzed in the C3 and C4 channels, respectively,

$$W(j, k) = \int_R f(x) \frac{1}{\sqrt{j}} \psi\left(\frac{x-k}{j}\right) dx \quad (1)$$

where $\frac{1}{\sqrt{j}} \psi\left(\frac{x-k}{j}\right)$ are the dilated and translated

versions of the wavelet function $\psi(x)$ at scale j and shift k , and $W(j, k)$ represents the CWT of the EEG signal $f(x)$. $W(j, k)$ is regarded as a 2D time-scale plot that retains the scale separation of ERP components [4]. However, the 2D time-scale plot $W(j, k)$ generated from the CWT is so noisy that it is difficult to accurately locate the active segment. The 2D anisotropic Gaussian filter is proposed to further enhance the active segment selection by smoothing the power spectrum and reducing the noise effects in the time-frequency domain. In addition, the best discriminant region is not situated at a single point but lies in a range of time and scale. We consider the time

and scale range by convoluting the above-mentioned 2D anisotropic Gaussian filter, and find that by using a wider frequency range a higher classification accuracy can be achieved from the acquired EEG signals than with a narrower one [15]. It is because the wider range contains all the mu and beta rhythmic components that are important for mental task classification. Therefore, we accumulate the power spectrum along the scale direction within the wide frequency range. The resulting profile represents the brain ERP responses for left and right EEG data. Student's two-sample t-statistics between the left and right data are evaluated from the training data set in both *C3* and *C4* channels,

$$t(k) = \frac{|\mu_L(k) - \mu_R(k)|}{\sqrt{\left(\frac{(N_L - 1) \cdot \sigma_L^2(k) + (N_R - 1) \cdot \sigma_R^2(k)}{N_L + N_R - 2} \right) \left(\frac{1}{N_L} + \frac{1}{N_R} \right)}} \quad (2)$$

where L and R represent left and right EEG data, respectively, N_s denotes the number of trials in state s , and $\mu_s(k)$ and $\sigma_s^2(k)$ stand for the mean and variance, respectively, of the profile in state s from the training data set. The denominator in equation (2) indicates the pooled variance of left and right data. $t(k)$ with different time k can form a 1D profile with respect to time, but it contains different characteristics. The peaks in the profile $t(k)$ imply that there are local maximal differences between left and right data in the time-scale domain, meaning the left and right EEG data are best discriminated at a particular time and scale range. The t-statistics are used to select the active segment. The active segment of length 1-s is then selected, with its center being the peak after the *C3* and *C4* channels are concatenated.

2.3. Feature extraction

Feature extraction is performed on the selected 1-s active segment rather than directly classifying the native EEG data without feature extraction. It greatly affects the performance of classification. In this study, the active segments are first band-pass filtered to the wide frequency range that contains all mu and beta rhythmic components using a Butterworth band-pass filter. The modified fractal dimension is then performed on wavelet data, which are transformed from filtered active segments.

It decomposes a signal into numerous details at various resolutions for multiresolution analysis, where each resolution represents a class of distinct physical characteristics within the signal. In other words, a

signal is characterized with the formulation by decomposing it into sub-bands, and each sub-band can be treated individually based on its characteristics. Multiresolution representation of the filtered active segments is achieved by discrete wavelet transform (DWT). The active segment A for each trial is represented in terms of the DWT as

$$A(x) = \sum_{k=-\infty}^{\infty} S_J(k) 2^{J/2} \phi(2^J x - k) + \sum_{j=1}^J \sum_{k=-\infty}^{\infty} D_j(k) 2^{j/2} \psi(2^j x - k) \quad (3)$$

where $S_J(k)$ and $D_j(k)$ represent the approximation and detail spaces of A , respectively, and $2^{J/2} \phi(2^J x - k)$ and $2^{j/2} \psi(2^j x - k)$ denote the dilated and translated versions of the scaling function $\phi(x)$ and wavelet function $\psi(x)$, respectively. The active segment A is then decomposed into individual subbands S_J , D_J , ..., and D_1 .

Fractal dimension (FD) is a popular method to extract fractal features. It is relatively insensitive to signal scaling and shows a strong correlation with human judgment of surface roughness [16]. The differential box counting (DBC) method covers a wide dynamic range with a low computational complexity. However, there are two serious shortcomings in the original DBC method, which make the estimation of fractal dimension inaccurate. We have proposed a modified method, namely modified fractal dimension, in our previous work [12]. First, the original DBC method is based on the difference between the minimum and maximum rectangle numbers, and is easily disturbed by noise. As the standard deviation of the amplitude represents the dispersion of the signal, we use the standard deviation to replace the difference in rectangle numbers in the original DBC method. Second, DBC produces a stair-like function and may result in the underestimation of fractal dimension. We overcome this problem by means of the floating calculation of fractal dimension.

The multiresolution fractal feature vectors (MFFVs) are formed by concatenating various-scale fractal features, calculated from the active segment itself and all of its subbands using the modified fractal dimension. The MFFVs reflect the roughness of the MI data at multiresolution.

2.4. Classification

It can be difficult for conventional neural networks to establish since neural networks usually need to

choose appropriate number of hidden layers and neurons to approximate the function in question to the desired accuracy. If the number in the network is too many, it may over-fit the training data and results in very poor generalization. The SVM pioneered by Vapnik [17] not only has a very steady theory in statistical learning, but guarantees to obtain the optimal decision function from a set of training data. The advantage of SVM is to balance the accuracy and generalization by maximizing the performance of network as well as minimizing the complexity of learning machine at the same time. The main idea of SVM is to construct a hyperplane as the decision surface in such a way that the margin of separation between positive and negative examples is maximized. The SVM optimization problem is

$$\begin{aligned} \min_w \quad & \frac{1}{2} w^T w + C \sum_{i=1}^N \xi_i \\ \text{subject to} \quad & \xi_i \geq 0, \forall i, \\ \text{and} \quad & d_i(w^T x_i + b) \geq 1 - \xi_i, \forall i = 1, 2, \dots, N \end{aligned} \quad (4)$$

where $g(x) = w^T x + b$ represents the hyperplane, w is the weighting vector, b is the bias term, x is the training vector with label d , C is the weighting constant, and ξ is the slack variable. It is then transformed into a convex quadratic dual problem. The discriminant function with optimal w and b , $g(x) = w_o^T x + b_o$, posterior to the optimization form becomes

$$g(x) = \sum_{i=1}^N \alpha_i d_i K(x, x_i) + b_o \quad (5)$$

where α is a Lagrange multiplier and $K(x, x_i)$ is a kernel function. Generally, appropriate kernel functions [17] are the polynomial kernel function $K(x_i, x_j) = (x_i^T x_j + 1)^p$ and the radial basis function (RBF) kernel function

$$K(x_i, x_j) = \exp\left(-1/2\sigma^2 \|x_i - x_j\|^2\right).$$

3. Results

3.1. Results of enhanced active segment selection

Fig. 2 shows the results of original and enhanced active segment selection. More specifically, the result of original active segment selection is shown in Fig. 2(a), whereas Fig. 2(b) show the results of enhanced active segment selection. For each subfigure, the first two plots represent 2D time-scale plots at channels C3

and C4 prior to scale accumulation, and the last plot displays the result of active segment selection at both C3 and C4 channels.

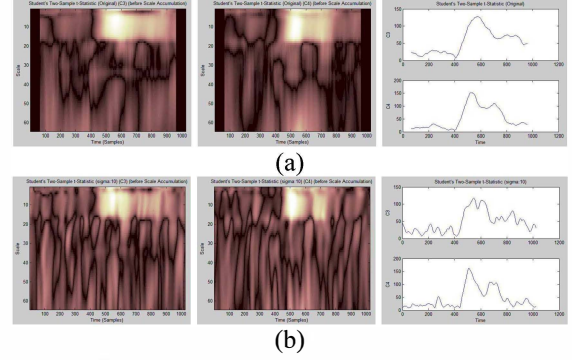


Fig. 2. Results of active segment selection. (a) For original active segment selection (b) For enhanced active segment selection.

3.2. Performance of enhanced active segment selection

The five-fold cross validation is carried out for classification tests of EEG data in this study. In other words, the dataset for each subject is divided into five subsets, and the following procedure is repeated five times. Each time, one of the five subsets is used as the test set and the other four are used as training set. The average recognition rate is immediately evaluated across all five folds.

To verify the efficacy of enhanced active segment selection, we compare the classification accuracy of SVM classifier for MFFV features among without, original active segment selection, and enhanced active segment selection at different σ s. The features are extracted from the event-related window for the case of “without active segment selection” in this experiment. The comparison results of classification accuracy are listed in Table 1. The results show that the average accuracy for original active segment selection increases to 80.8%. In addition, the average accuracy can further raise to 83.6% after enhanced active segment selection is used. Moreover, we observe that enhanced active segment selection gives the best results for all subjects. Therefore, the results indicate that enhanced active segment selection can further improve the performance of MI classification.

3.3. Performance between LDA and SVM classifiers

Finally, an experiment is performed to estimate the performance of SVM classifier. Table 2 lists the comparisons of classification accuracy under enhanced active segment selection and MFFV features between LDA and SVM classifiers. These two classifiers use the same active segments. That is, the listed results only demonstrate the discrepancy of performance between different classifiers. The results show that the average accuracy for LDA is 79.5%, whereas that for SVM is 83.6%. We observe that classification with SVM obtains the best results for all subjects. Accordingly, the results denote that the SVM classifier is better than the LDA one in EEG classification.

Table 1

Comparison of classification accuracy between original and enhanced active segment selection

Classification Accuracy	Dataset			Average
	S1	S2	S3	
Original ASS	83.8%	76.7%	81.9%	80.8%
Enhanced ASS	86.7%	79.6%	84.6%	83.6%

Table 2

Comparison of classification accuracy between LDA and SVM

Classification Accuracy		LDA	SVM
Dataset	S1	81.6%	86.7%
	S2	75.1%	79.6%
	S3	82.0%	84.6%
Average		79.5%	83.6%

3.4. Two-way ANOVA and multiple comparison tests

In addition, two-way analysis of variance (ANOVA) and multiple comparison tests are performed in the experiments. The statistical analyses with two-way ANOVA are used to evaluate that the difference is significant or not for the two factors, methods and subjects. After analyzing with the two-way ANOVA, multiple comparison tests are used to estimate the p -values and significance of each pair of methods. The results of tests will be discussed in detail in the next section.

4. Discussion

4.1. Statistical evaluation of enhanced active segment selection

Table 1 lists the comparisons of performance between original and enhanced active segment selection under the MFFV features and SVM classifier. In addition, two-way ANOVA and multiple comparison tests are performed to verify if original and enhanced active segment selection is significantly different or not. The results indicate that enhanced active segment selection is significantly better than original active segment selection (p -value 0.0031). In summary, enhanced active segment selection can further improve the accuracy by refining the selection of essential information from event-related windows.

4.2. Statistical evaluation between LDA and SVM classifiers

Table 2 lists the comparisons of performance between the LDA and SVM classifiers under the enhanced active segment selection and MFFV features. Moreover, two-way ANOVA and multiple comparison tests are performed again to verify if these two classifiers are significantly different. The results reveal that there is significant difference between LDA and SVM (p -value 0.0019). Hence, the SVM is a better classifier in BCI applications.

5. Conclusion

In this study, we have proposed an EEG analysis system for the single-trial MI classification. Combined with enhanced active segment selection and multiresolution fractal features, the SVM is applied for the discriminant of EEG data. The enhanced active segment selection further refines the location of active segments to substantially reduce the length of event-related window to 1-s segment and increase classification accuracy at the same time. The MFFVs are then extracted from wavelet data by modified fractal dimension. Finally, the SVM is used for classification. The experimental results demonstrate that enhanced active segment selection can further improve the performance, and the MFFVs and SVM are satisfactory features and classifier in the applications of BCI work.

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